SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2, BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 Total Marks: 30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I **B Tharun Kumar / 192424356**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B Tharun Kumar

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

Answer:

Step 1: Understanding the Problem and Constraints

The branch office is located in a remote rural area where wired broadband (fibre/DSL) is not available. The design must satisfy four competing constraints simultaneously: bandwidth adequacy for cloud applications and video conferencing, reliability for uninterrupted operations, low cost due to a limited connectivity budget, and scalability so the link can grow as the branch expands. These constraints are interdependent — increasing reliability or bandwidth usually increases cost, so the solution must be evaluated using quantitative trade-off formulas rather than guesswork.

Step 2: Estimating Required Bandwidth (Constraint Modelling)

The first constraint to quantify is bandwidth. The total required bandwidth can be modelled as:

B_required = Σ(N_app × R_app) + O_overhead

where N_app = number of concurrent sessions of an application,

R_app = bit-rate required per session, and

O_overhead = protocol/encryption overhead (typically 10–20% of payload).

Applying typical enterprise figures — HD video conferencing needs approximately 1.5–2 Mbps per stream (up/down), and cloud application access (SaaS, ERP, email sync) needs roughly 0.5–1 Mbps per active user — for a small branch of about 15–20 users with 4–5 concurrent video calls:

$$B\_required = (5 \times 2\ Mbps) + (20 \times 0.75\ Mbps) + 15\%\ overhead$$

$$B\_required \approx (10 + 15) \times 1.15 \approx 28.75\ Mbps\ (\approx 30\ Mbps\ symmetrical)$$

This calculated figure, not an arbitrary number, becomes the sizing target for the WAN link, and it also defines the scalability margin needed for future growth.

Step 3: Selecting the Internet Connectivity Option

Since wired fibre/DSL is unavailable, the two feasible last-mile technologies are satellite broadband (e.g., VSAT/LEO satellite) and cellular broadband (4G/5G fixed-wireless access). The choice is made using a cost-reliability-bandwidth trade-off ratio:

$$Value\ Index\ (VI) = (Available\ Bandwidth \times Reliability\ \%) / Monthly\ Cost$$

Higher VI = better constraint satisfaction per rupee spent.

4G/5G fixed-wireless typically offers 20–100 Mbps at lower monthly cost with 95–98% uptime, giving a higher VI than traditional VSAT (which offers stable but expensive and higher-latency bandwidth, ~500–700 ms). Since video conferencing is latency-sensitive (ITU-T G.114 recommends one-way latency below 150 ms for acceptable voice/video quality), cellular broadband is preferred as the primary link, with low-cost VSAT or a second cellular carrier retained only as a backup — because relying on a single ISP violates the reliability constraint.

Step 4: Reliability Through Redundancy (Constraint Satisfaction)

Uninterrupted operation is a hard constraint, so the design must avoid a single point of failure. Reliability of a dual-link (failover) configuration is computed using the parallel-redundancy formula:

$$R\_system = 1 - [(1 - R1) \times (1 - R2)]$$

If R1 (primary 4G/5G link) = 0.97 and R2 (backup VSAT/second SIM) = 0.90:

$R\_system = 1 - [(0.03) \times (0.10)] = 1 - 0.003 = 0.997$ *(≈ 99.7% uptime)*

This shows mathematically that adding a modest secondary link raises effective availability from ~97% to ~99.7%, satisfying the 'uninterrupted operations' constraint far more efficiently than upgrading a single link, which is the cost-optimal path.

Step 5: Networking Devices Required

- SD-WAN capable router / dual-WAN router — automatically load-balances and fails over between the two internet links.
- Firewall / UTM appliance (Unified Threat Management) — provides stateful inspection, intrusion prevention, and content filtering at the branch edge.
- Managed Layer-2/3 switch — for internal LAN segmentation and future scalability of ports.
- VPN concentrator (or firewall with built-in IPsec/SSL VPN) — establishes a secure, encrypted tunnel to headquarters.
- Wireless Access Point — for flexible in-office connectivity without additional structured cabling cost.

Step 6: Security Mechanism for Secure HQ Communication

Secure communication with headquarters is achieved through a site-to-site IPsec VPN tunnel, which uses IKE (Internet Key Exchange) for key negotiation and AES-256 for payload encryption. The effective throughput available to applications after VPN encapsulation overhead is:

$B\_usable = B\_link \times (1 - VPN\_overhead\_\%)$

For IPsec/AES-256, overhead ≈ 8–10%, so:

$B\_usable = 30 \text{ Mbps} \times (1 - 0.10) = 27 \text{ Mbps}$

This confirms 27 Mbps of usable capacity still exceeds the 28.75 Mbps raw requirement only marginally, so the link should be provisioned with an additional 20–25% headroom, giving a recommended purchased bandwidth of ~35–40 Mbps to remain within safe operating margins.

Step 7: Cost Justification

Total Cost of Ownership (TCO) is evaluated over a planning horizon using:

$$TCO = C_{capex} + (C_{opex} \times T)$$

where C_{capex} = one-time device cost, C_{opex} = monthly recurring (ISP + VPN license), T = months of use.

Because fixed-wireless 4G/5G avoids the trenching and installation costs of fibre in a rural location, C_capex is minimal, and the dual-link redundancy adds only a fraction of the primary link's cost, keeping TCO low while still satisfying the reliability formula derived above.

Step 8: Scalability

Future scalability is planned using a linear growth projection:

$$B_{future}(\text{year } n) = B_{required} \times (1 + g)^n$$

where g = expected annual growth in users/applications (commonly 15–20% for expanding branches).

Applying g = 15% over 3 years: $B_{future} = 30 \times (1.15)^3 \approx 45.6$ Mbps — indicating the chosen SD-WAN router and cellular ISP plan must support tiered upgrades up to ~50 Mbps without hardware replacement, which fixed-wireless/5G plans allow simply by changing the subscribed data tier.

Conclusion

The recommended architecture is a dual-WAN (primary 4G/5G + backup cellular/VSAT) fixed-wireless connection behind an SD-WAN router with an integrated UTM firewall and site-to-site IPsec VPN to headquarters. This design is derived directly from the bandwidth-estimation, redundancy-reliability, cost-index, and growth-projection formulas above, and it satisfies all four constraints — bandwidth, reliability, cost, and scalability — without requiring wired infrastructure that is unavailable at the site.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Answer:

Step 1: Understanding the Problem and Constraints

The core constraints are: (a) capacity — more than 3,000 concurrent users, (b) coverage — classrooms, labs, libraries and hostels of varying size and wall density, (c) minimal interference between neighbouring access points, (d) security for authentication, and (e) a limited AP budget. As with Question 1, the design is derived by first quantifying capacity and coverage needs mathematically, then optimizing AP count against the budget constraint.

Step 2: Estimating the Number of Access Points Needed (Capacity Constraint)

Each Wi-Fi access point has a practical concurrent-client capacity, not just a coverage radius. The number of APs required by capacity is:

N_AP(capacity) = N_users / C_AP

where C_AP = practical concurrent users an enterprise AP can serve with acceptable performance (typically 50–60 for dense classroom use on modern 802.11ac/ax APs).

For 3,000 students, assuming a realistic concurrent-usage factor of 60% (not all students are online at the same instant):

N_concurrent = 3000 × 0.6 = 1800 users

N_AP(capacity) = 1800 / 55 ≈ 33 access points (minimum by capacity alone)

Step 3: Estimating APs Needed by Coverage (Area Constraint)

Coverage-driven AP count uses the free-space/indoor coverage radius formula. For indoor enterprise APs with typical usable radius $r \approx 20–25$ m (accounting for classroom walls), the coverage area per AP is:

$$A\_AP = \pi \times r^2 = \pi \times (20)^2 \approx 1257\ m^2$$

$$N\_AP(coverage) = Total\_Campus\_Area / A\_AP$$

For a mid-sized campus of roughly 40,000 m^2 of usable indoor teaching/hostel space: N_AP(coverage) = 40,000 / 1,257 ≈ 32 access points. The final AP count is taken as the maximum of the two estimates (capacity governs, since it is higher in dense zones such as libraries and lecture halls):

$$N\_AP(final) = MAX[N\_AP(capacity), N\_AP(coverage)] \approx 33–35\ access\ points$$

This gives the university a data-justified AP budget instead of an arbitrary number, directly addressing the 'limited number of access points' constraint.

Step 4: Access Point Placement Strategy

- Weight AP density toward high-occupancy zones (libraries, large lecture halls, hostels) using the capacity formula above rather than distributing APs uniformly by floor area.
- Mount APs on ceilings at a height of 3–4 m with omnidirectional antennas in open areas, and directional antennas in long corridors, to maximize the effective coverage-per-AP ratio (A_AP).
- Stagger AP placement in a honeycomb (hexagonal) pattern rather than a grid, since hexagonal packing minimizes the number of APs needed to cover a given area without coverage gaps — the packing efficiency of hexagonal cells is mathematically higher than square grids for equal radius r.
- In hostels, place one AP per 2–3 rooms/floor segment to balance wall-attenuation losses (concrete walls typically attenuate Wi-Fi signal by 6–10 dB per wall).

Step 5: Frequency Band Selection and Interference Minimization

To minimize co-channel interference, the design uses dual-band operation with non-overlapping channel allocation. In the 2.4 GHz band only channels 1, 6, and 11 are non-overlapping (each 20 MHz wide with 5 MHz spacing, so the number of truly non-overlapping channels is):

$$N\_nonoverlap(2.4GHz) = \lfloor(Total\_BW - Channel\_BW) / Spacing\rfloor + 1 = \lfloor(83.5 - 20)/5\rfloor + 1 \approx 3 \text{ channels}$$

Because only 3 non-overlapping channels exist at 2.4 GHz, adjacent APs reusing the same channel would cause co-channel interference. The 5 GHz band offers up to 24 non-overlapping 20 MHz channels, so high-density zones (libraries, lecture halls) should primarily use 5 GHz/6 GHz (Wi-Fi 6/6E) for capacity and reduced interference, while 2.4 GHz is reserved only for range-extension in low-density hostel corridors. APs are then assigned channels using a reuse pattern such that no two adjacent cells share a channel, following the same principle as cellular frequency-reuse planning:

Co-channel Reuse Distance: $D = r \times \sqrt{3K}$

where K = frequency reuse factor (e.g., $K=3$ for a 3-channel plan). A larger D relative to r ensures adjacent same-channel APs are spaced far enough apart to avoid interference.

Step 6: Authentication and Security Method

Given that thousands of students share the network, the recommended authentication is WPA3-Enterprise using IEEE 802.1X with a RADIUS/AAA server (or EAP-TLS/PEAP), so each student authenticates with individual university credentials rather than a shared pre-shared key. This provides per-user encryption keys, centralized access revocation for lost devices, and reduces the load-balancing/security risk of a single shared password on 3,000+ devices. A captive portal with 802.1X can additionally be used for guest access in libraries with restricted bandwidth policies enforced via QoS (Quality of Service) tagging.

Step 7: Balancing Coverage, Performance, Security and Cost

The final AP count (33–35) is the minimum that satisfies both the coverage formula and the capacity formula simultaneously — deploying fewer APs would violate the capacity constraint (queuing delay for users), while deploying far more would exceed the budget without proportional benefit, since coverage overlap beyond ~15–20% offers diminishing returns. This can be expressed as an optimization objective:

Minimize: Cost = N_AP × Cost_per_AP

Subject to: N_AP ≥ N_AP(capacity) AND N_AP ≥ N_AP(coverage)

Security (WPA3-Enterprise/802.1X) and interference control (dual-band channel reuse planning) are layered on top of this minimum AP count at negligible additional hardware cost, since they are configuration-based rather than requiring extra devices — meaning all four constraints (coverage, performance, security, cost) are satisfied within the same AP budget.

Conclusion

The proposed design uses approximately 33–35 enterprise dual-band access points placed in a hexagonal pattern weighted toward high-density zones, using 5 GHz/6 GHz as the primary capacity band with a 3-channel reuse plan on 2.4 GHz for range extension, and WPA3-Enterprise with 802.1X/RADIUS authentication for secure, individually-revocable access — all derived from the capacity, coverage, and channel-reuse formulas above rather than an arbitrary deployment.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Answer:

Step 1: Understanding the Problem and Constraints

A financial institution's network must satisfy strict, sometimes conflicting constraints: security (protection of customer data, regulatory frameworks such as PCI-DSS/RBI/ISO 27001), performance (transaction processing must not be slowed), compliance (mandatory logging, segmentation, and audit trails), and budget (no significant increase in operational cost). The design approach is to quantify the trade-off between security depth and performance overhead using layered defense formulas rather than adding security devices indiscriminately.

Step 2: Network Segmentation Strategy (Subnetting/VLAN Design)

Segmentation limits the blast radius of a breach and is mandatory for PCI-DSS compliance. The institution's network is divided into VLANs/subnets by trust zone: (1) Public-facing DMZ (online banking web/app servers), (2) Core banking / database zone (sensitive customer data), (3) Employee/internal LAN, (4) Management/IDS zone. Subnet sizing uses the standard host-capacity formula:

$$\text{Usable Hosts} = 2^{(32 - \text{Prefix})} - 2$$

$$\text{e.g., a /26 subnet: } 2^{(32-26)} - 2 = 2^6 - 2 = 62 \text{ usable hosts}$$

Each VLAN is sized with the smallest prefix that satisfies its host count plus ~20% growth headroom, computed as:

$$Required\ Prefix\ Hosts = N\_devices \times (1 + growth\_\%)$$

then choose smallest $2^n - 2 \geq Required\ Prefix\ Hosts$

This avoids wasting IP space (cost/administrative overhead) while leaving room for expansion — satisfying both the compliance requirement for isolation and the budget constraint of efficient address planning.

Step 3: Firewall Deployment (Defense-in-Depth)

A two-tier firewall architecture is used: an external (perimeter) firewall between the Internet and the DMZ, and an internal firewall between the DMZ and the core banking zone — creating a 'defense-in-depth' architecture rather than a single point of inspection. The probability that an attacker breaches both layers is the product of individual breach probabilities:

$$P\_breach(total) = P\_breach(perimeter) \times P\_breach(internal)$$

e.g., if each firewall independently has a 5% chance of being bypassed (P=0.05):

$$P\_breach(total) = 0.05 \times 0.05 = 0.0025\ (0.25\%)$$

This shows mathematically that layering two moderately strong firewalls is far more effective than one very expensive 'best-in-class' firewall, satisfying the security constraint while controlling cost. Firewall throughput must also be sized to avoid becoming a performance bottleneck, using:

$$Required\ Firewall\ Throughput \geq Peak\_Concurrent\_Transactions \times Avg\_Packet\_Size \times 8 / Time\_Window$$

ensuring inspected (stateful + IPS) throughput still exceeds peak banking-hour traffic.

Step 4: Authentication Methods

Employee access to sensitive systems uses Multi-Factor Authentication (MFA) combined with Role-Based Access Control (RBAC), so each employee's access is scoped only to the data required for their role (principle of least privilege). The security strength improvement from adding a second authentication factor is modelled as:

$P\_compromise(MFA) = P\_compromise(factor1) \times P\_compromise(factor2)$

e.g., password compromise P=0.1 and OTP/token compromise P=0.02:

$P\_compromise(MFA) = 0.1 \times 0.02 = 0.002$ *(0.2%), a 50x improvement over single-factor*

For customer-facing online banking, authentication additionally uses TLS 1.3 for transport encryption and OTP-based two-factor login, satisfying regulatory requirements for strong customer authentication without materially increasing latency, since MFA validation typically adds only tens of milliseconds relative to total transaction time.

Step 5: Intrusion Detection/Prevention System (IDS/IPS)

A hybrid IDS is deployed: signature-based detection at the perimeter (fast, low false-positive rate for known attacks) and anomaly-based detection inside the core banking zone (to catch zero-day/insider threats). IDS effectiveness is evaluated using detection-rate and false-positive trade-off formulas:

$$\text{Detection Rate (Sensitivity)} = TP / (TP + FN)$$

$$\text{False Positive Rate} = FP / (FP + TN)$$

$$\text{Precision} = TP / (TP + FP)$$

The IDS threshold is tuned to maximize detection rate while keeping the false-positive rate low enough that security staff are not overwhelmed with false alerts (an operational cost). This tuning directly balances the security and performance/cost constraints: a threshold that is too sensitive raises FP and burdens staff (cost), while one that is too relaxed reduces TP (security risk).

Step 6: Performance Impact Analysis

The cumulative latency added by segmentation + firewall inspection + MFA + IDS must remain within the acceptable transaction-response budget. This is modelled as a serial latency chain:

$$T\_total = T\_network + T\_firewall + T\_MFA + T\_IDS + T\_application$$

Constraint: T_total ≤ T_SLA (e.g., 2 seconds for an online banking transaction)

Each security control is selected/sized (e.g., hardware-accelerated firewalls, inline IDS with dedicated ASICs) such that the sum of its added latency remains a small fraction of T_SLA, ensuring compliance-mandated controls do not violate the performance constraint.

Step 7: Cost Justification

Return on Security Investment (ROSI) is used to justify each control against the operational-cost constraint:

ROSI = (Risk_Mitigated_Value − Cost_of_Control) / Cost_of_Control

A control is justified when ROSI > 0, i.e., the reduction in expected breach loss exceeds its cost.

Because segmentation, dual-firewalls, MFA and hybrid IDS all reduce breach probability multiplicatively (as shown in Steps 3–4), the marginal risk reduction per rupee spent is high, making this layered design cost-justified compared to relying on a single expensive perimeter appliance.

Conclusion

The proposed architecture segments the network into DMZ, core banking, internal LAN, and management VLANs using capacity-based subnetting; deploys a two-tier (perimeter + internal) firewall to achieve multiplicative breach-probability reduction; enforces MFA with RBAC for authentication; and uses a hybrid signature/anomaly-based IDS tuned via detection-rate/false-positive formulas. Latency-budget and ROSI calculations confirm the design meets performance and cost constraints while satisfying regulatory compliance and strong security posture.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand